

# How Predictable Are Rare Precipitation Extremes? A Calibration-Aware Assessment for London

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## Abstract

How predictable are rare precipitation extremes, and how should that predictability be measured? We address both questions with thirty years (1989–2018) of a standardised daily precipitation-extreme index for London, augmented with strictly antecedent ERA5 atmospheric drivers. Methodologically, we argue that rare-event forecasts must be judged by *calibration-aware proper scores* relative to climatology — not by discrimination metrics or by conformal coverage, which on its own can give a misleading impression of forecast quality — and we assemble an evaluation protocol (Brier reliability/resolution decomposition, threshold- and quantile-weighted CRPS, skill scores, bootstrap confidence intervals, and multi-split robustness) that guards against the single-seed and quantile-selection pitfalls that inflate apparent skill. Empirically, across a bake-off of models from logistic regression to a deep Gaussian-process extreme-value hurdle, we find that (i) **simple calibrated models outperform the elaborate ones**, and the choice of response distribution dominates model class; (ii) **occurrence** of an extreme is weakly but genuinely predictable from antecedent state (led by convective-instability and circulation indices); and (iii) **the magnitude of the largest events** carries weak but robust predictability tied to antecedent convective instability (CAPE/CIN), with a continuous-ranked-probability skill over climatology whose bootstrap interval excludes zero across four independent train/test splits. The effects are small — precipitation extremes are intrinsically hard — but real, physically grounded, and rigorously established.

**Keywords:** predictability; extreme-value theory; proper scoring rules; calibration; precipitation extremes; convective instability.

## 1 Introduction

Extreme precipitation is among the most damaging natural hazards facing temperate cities. Short, intense rainfall episodes trigger surface-water and river flooding, disrupt transport, overwhelm drainage, and impose large and unevenly distributed economic and social costs. In London such events occur on only a small fraction of days, yet a single episode can cause disproportionate harm, and both observational records and climate projections indicate that the frequency and intensity of heavy rainfall are increasing (?). Skilful and, above all, *trustworthy* early warning of these events would let emergency managers, infrastructure operators, and insurers act pre-emptively. This motivates probabilistic forecasts of extreme precipitation at horizons of days to weeks ahead.

The statistical obstacles are severe. Extreme days are rare — typically 1–5% of the record — so a model that simply predicts “no extreme” is accurate yet useless, and conventional accuracy is the wrong yardstick. The daily precipitation index is zero-inflated, strongly right-skewed, and heavy-tailed, violating the Gaussian assumptions underlying standard regression. The observational record is short relative to the rarity of the events, leaving few positive examples from which to learn, and the day-to-day persistence of extremes is limited, so intrinsic predictability is modest.

Together these features cap raw forecast *skill*, a ceiling any honest treatment must confront rather than obscure.

We argue that, precisely because raw predictability is limited, the decisive quality of a forecast in this setting is not discrimination but *calibration*: whether the issued probabilities can be trusted. A forecast that assigns a 5% probability to an extreme is useful for decision-making only if extremes do occur on about 5% of such days; an overconfident forecast that systematically overstates rare-event probabilities causes false alarms and wasted resources, while an underconfident one breeds complacency. The appropriate goal is therefore to *maximise sharpness subject to calibration* (?): to be as informative as the data allow while remaining honest. Calibration matters most exactly where it is hardest to achieve — in the tail — and evaluating it there is itself subtle, as the *forecaster’s dilemma* makes clear (?).

No existing methodology delivers calibrated, guaranteed tail probabilities for rare-event forecasting, because the three relevant tools each supply only part of the answer. *Extreme-value theory* (EVT) furnishes asymptotically justified tail models, but is most often used descriptively — for return levels or non-stationary tail regression — rather than for flexible, covariate-driven probabilistic forecasting, and it offers no finite-sample coverage guarantee (??). *Flexible machine learning*, including deep models and Gaussian processes, captures non-linear dependence on atmospheric predictors and yields predictive distributions, but those distributions are typically miscalibrated in the tail and come with no guarantee (??). *Conformal prediction* provides distribution-free, finite-sample coverage, but its standard form under-covers the tail — the relevant high quantile cannot be resolved from a finite calibration set — and assumes exchangeability, which daily time series violate (??). Moreover, essentially all two-part (hurdle) forecasters assume that the *occurrence* of an extreme and its *intensity* are conditionally independent, discarding the abundant occurrence data that could inform the data-starved tail.

Rather than advocate a single model, we therefore ask a more basic pair of questions — *how predictable are these extremes, and how should that predictability be measured?* — and answer them with a careful evaluation protocol and a bake-off of models spanning logistic regression, zero-inflated distributional regressions (log-normal, gamma, and the threshold-free extended generalised Pareto distribution), and a deep Gaussian-process extreme-value hurdle with conformal calibration.

## Contributions.

1. A **calibration-aware evaluation protocol** for rare-event precipitation forecasting: Brier reliability/resolution decomposition, threshold- and quantile-weighted CRPS, skill scores against climatology, bootstrap confidence intervals, and multi-split robustness — designed to resist the single-seed and quantile-selection artefacts that inflate apparent skill, and to demonstrate that conformal coverage is insufficient as a stand-alone measure of forecast quality.
2. A **rigorous predictability assessment**: occurrence of an extreme is weakly but genuinely predictable from antecedent state; and the *magnitude* of the largest events carries weak but robust predictability tied to antecedent **convective instability** (CAPE/CIN), with a CRPS skill over climatology whose bootstrap interval excludes zero across four independent train/test splits.
3. A **cautionary methodological finding**: across the bake-off, *simpler calibrated models outperform the elaborate deep-GP-EVT model*, and the choice of response distribution dominates model class (common heavy-tailed choices are badly miscalibrated).
4. As a methodological by-product, a **dependent (coregionalised) hurdle** that borrows strength between occurrence and intensity, validated on synthetic data.

Empirically, we study the daily standardised precipitation index for London over 1989–2018. The 95th-percentile index (with 417 extreme days) is the primary target, as the rarer

99th-percentile index (137 events) is too data-scarce for the held-out intensity analysis (its occurrence-only results are in the Supplementary Material). Predictors are strictly antecedent ERA5 atmospheric drivers (circulation, moisture, and convective-instability fields) and large-scale teleconnection indices. Because the index is itself an ERA5-derived precipitation product, we exclude all precipitation fields from the predictors and lag every driver, so the assessment is of genuine antecedent predictability rather than diagnosis.

The remainder of the paper is organised as follows. Section ?? positions our work within the literatures on extreme-value modelling, Gaussian processes, conformal prediction, and forecast evaluation. Section ?? describes the data and predictors. Section ?? sets out the models we compare, the conformal calibration, and the evaluation protocol. Section ?? presents the results, and Sections ??–?? discuss implications and conclude.

## 2 Related work

Our work sits at the intersection of four literatures; we review each in turn and then state precisely how this study differs.

**Statistical modelling of precipitation extremes.** Extreme-value theory provides the asymptotic foundation for tail modelling. The peaks-over-threshold approach models exceedances above a high threshold by the Generalised Pareto distribution (??), the framework we adopt for the intensity component; the block-maxima approach uses the Generalised Extreme Value distribution (?). EVT is widely applied in hydrology and climatology (?), including non-stationary and spatial Bayesian formulations of precipitation extremes (?). This body of work is, however, predominantly *descriptive* — estimating return levels or letting tail parameters vary smoothly with a few covariates — rather than producing flexible, covariate-rich *forecasts* at a fixed lead time, and it offers no finite-sample coverage guarantee. We retain the GPD tail but embed it in a flexible predictive model and calibrate it conformally.

**Machine learning and deep learning for extremes.** Data-driven methods increasingly target extremes: ? forecast block maxima with a deep network constrained by GEV theory, and neural networks have been used to estimate extreme-value parameters. More broadly, predictive uncertainty in deep models is quantified with deep ensembles (?) and Monte-Carlo dropout (?). These methods capture non-linear dependence on predictors and return uncertainty estimates, but they are typically miscalibrated in the tail and lack distribution-free guarantees — the gap our conformal layer closes.

**Gaussian processes: depth and multiple outputs.** Gaussian processes offer principled non-parametric regression with built-in uncertainty (?), made scalable through sparse variational inference with inducing points (??). Expressiveness is increased by composing layers in deep Gaussian processes (?), trained by doubly-stochastic variational inference (?), or by learning the kernel through a neural feature map in deep kernel learning (?). Correlated outputs are modelled by multi-output GPs and coregionalisation (??). A GP with a Gaussian likelihood cannot represent heavy-tailed, skewed, zero-inflated responses; we therefore pair the GP prior with an EVT likelihood and use coregionalisation not to share information across tasks but to couple the two halves of a hurdle model.

**Two-part, hurdle, and selection models.** Semicontinuous and zero-inflated outcomes are commonly modelled by two-part (hurdle) decompositions separating occurrence from intensity. Allowing the two parts to be *dependent* has a long history: correlated random-effects formulations (?), copula couplings, and the sample-selection models of econometrics (?), which explicitly address the correlation between a selection (occurrence) equation and an outcome (intensity)

equation. To our knowledge this dependence has not been exploited within a deep-GP-EVT *forecasting* model, where — because intensity is observed only on the rare extreme days — borrowing strength from the abundant occurrence data is especially valuable.

**Conformal prediction.** Conformal prediction yields distribution-free, finite-sample coverage for any underlying predictor (?). Split-conformal methods make this practical for regression (?), and conformalised quantile regression adapts the intervals to heteroscedasticity (?). Exchangeability can be relaxed for covariate shift (?) and, crucially for us, for time series and online distribution shift through adaptive conformal inference (??). Coverage at very high confidence levels — the extreme regime — is addressed by ?, who combine EVT with conformal prediction to build reliable intervals for high-impact events. That work, the closest to ours, constructs *regression intervals* from a black-box extreme-quantile estimator under (weighted) exchangeability; it uses neither a Gaussian-process nor a generative model, does not forecast exceedance *probabilities*, and does not model temporal dependence natively. Here, by contrast, tail-targeted and temporally-adaptive conformal calibration is applied to a generative model, within a broader predictability study.

**Forecast evaluation of extremes.** Probabilistic forecasts are assessed with proper scoring rules (?) under the paradigm of maximising sharpness subject to calibration (?); the Brier score admits a reliability–resolution decomposition (?). For extremes, threshold- and quantile-weighted scoring rules focus evaluation on the tail while remaining proper (?), and naive restriction to extreme observations can paradoxically reward miscalibrated forecasts — the forecaster’s dilemma (?). Our evaluation is built on these principles.

**Positioning.** Much of this literature proposes ever more expressive models for extremes. We take the complementary stance of *measuring* how much is predictable and which modelling choices matter, under an evaluation protocol built from proper scoring rules, skill against climatology, and bootstrap and multi-split robustness. In doing so we find — against the prevailing direction — that under this dataset and evaluation framework simple calibrated models outperform an elaborate deep-GP extreme-value model, and that conformal coverage, increasingly reported as a calibration guarantee, is insufficient on its own as a measure of forecast quality. This emphasis on rigorous predictability assessment, rather than a new architecture, is the gap we fill.

## 3 Data

### 3.1 The London precipitation-extreme index

Our response variable is a standardised daily precipitation-extreme index for London derived from the Copernicus Climate Data Store product *Extreme precipitation risk indicators for Europe and European cities (1950–2019)*. The underlying field is the ERA5 reanalysis dynamically downscaled to a  $2\text{ km} \times 2\text{ km}$  grid (ERA5-2km) with urban parameterisation, which resolves London as 53 grid cells. For each cell and day the indicator is the standardised exceedance

$$\mathcal{I} = \frac{P}{P_q}, \quad (1)$$

where  $P$  is daily precipitation and  $P_q$  is the fixed  $q$ th percentile of wet days ( $P \geq 1\text{ mm}$ ) over the 1950–2019 reference period. The index is dimensionless, and  $\mathcal{I} > 1$  marks an extreme: precipitation exceeding the historical  $q$ th-percentile level. We study both the  $q = 95$  and  $q = 99$  thresholds. To obtain a single city-level series we take the daily maximum of  $\mathcal{I}$  over the 53 cells, so an extreme day is one on which *any* part of London exceeds its threshold. We analyse the thirty-year record 1989–2018, extending the ten-year (2009–2018) record used in preliminary work by a factor of three to enlarge the scarce set of extreme examples.

Table 1: Summary characteristics of the London precipitation-extreme index (1989–2018;  $N = 10,957$  days), by standardisation threshold. Extreme days are those with  $\mathcal{I} > 1$ .

|                           | 95th percentile | 99th percentile |
|---------------------------|-----------------|-----------------|
| Extreme days (total)      | 417             | 137             |
| Extreme rate (%)          | 3.81            | 1.25            |
| train 1989–2008 (7,305 d) | 287 (3.93%)     | 101 (1.38%)     |
| test 2009–2018 (3,652 d)  | 130 (3.56%)     | 36 (0.99%)      |
| Days equal to zero (%)    | 25.5            |                 |
| Median index              | 0.016           | 0.010           |
| Maximum index             | 6.46            | 3.95            |
| Peak season               | July–August     |                 |

### 3.2 Exploratory characteristics

The series comprises 10,957 daily observations. Its defining features — which directly motivate the modelling choices of Section ?? — are summarised in Table ?? and Figure ?. First, extremes are *rare*: 3.8% of days at the 95th percentile and only 1.3% at the 99th, the latter giving just 137 extreme days in thirty years. Second, the index is *zero-inflated and heavy-tailed*: about a quarter of days are exactly zero, the median is below 0.02, yet the maximum exceeds 6 at the 95th-percentile standardisation — a strong right skew incompatible with a Gaussian response. Third, occurrence is *seasonal*: the monthly extreme rate peaks in summer (July–August, reaching 6% at the 95th percentile and 3% at the 99th) and is lowest in late winter and early spring, consistent with convective summer rainfall. Fourth, the extreme indicator exhibits only *limited temporal autocorrelation*, so the persistence available to an autoregressive forecaster is modest — one reason that drivers beyond the index’s own history (Section ??) are essential.

### 3.3 Antecedent predictors

All predictors are *strictly antecedent*: for a horizon- $h$  forecast of day  $t + h$  they are functions only of information available up to day  $t$ . We use three groups.

1. **Index history.** Lagged values of the index (up to 14 days), rolling means and maxima over 7/14/30-day windows, counts of recent exceedances, and harmonic encodings of day-of-year and month to capture seasonality.
2. **ERA5 atmospheric drivers.** Antecedent fields from the ERA5 reanalysis over a synoptic-scale window around London: mean sea-level pressure and its gradients, geopotential height (500/850 hPa), specific and total-column water vapour, integrated water-vapour transport, near-surface temperature and winds, and convective-instability diagnostics — convective available potential energy and inhibition, the K-index, and the Total-Totals index. Fields are aggregated to daily statistics (means, and maxima for the convective indices) and reduced to regional summaries.
3. **Teleconnection indices.** Lagged North Atlantic Oscillation, Arctic Oscillation, and an El Niño–Southern Oscillation index, capturing large-scale circulation states relevant at longer horizons.

In the analysis the drivers are introduced *cumulatively* — index history; + teleconnections; + ERA5 circulation and moisture; + convective instability — so that each increment isolates a class of predictors. All features are standardised using training-set statistics only; the complete list and the retrieval recipe are in Appendix ??.

**Remark 1** (No leakage; perfect-prognosis framing). *ERA5 is a reanalysis that assimilates observations, so using fields at or after the target day would be circular. We therefore use only*

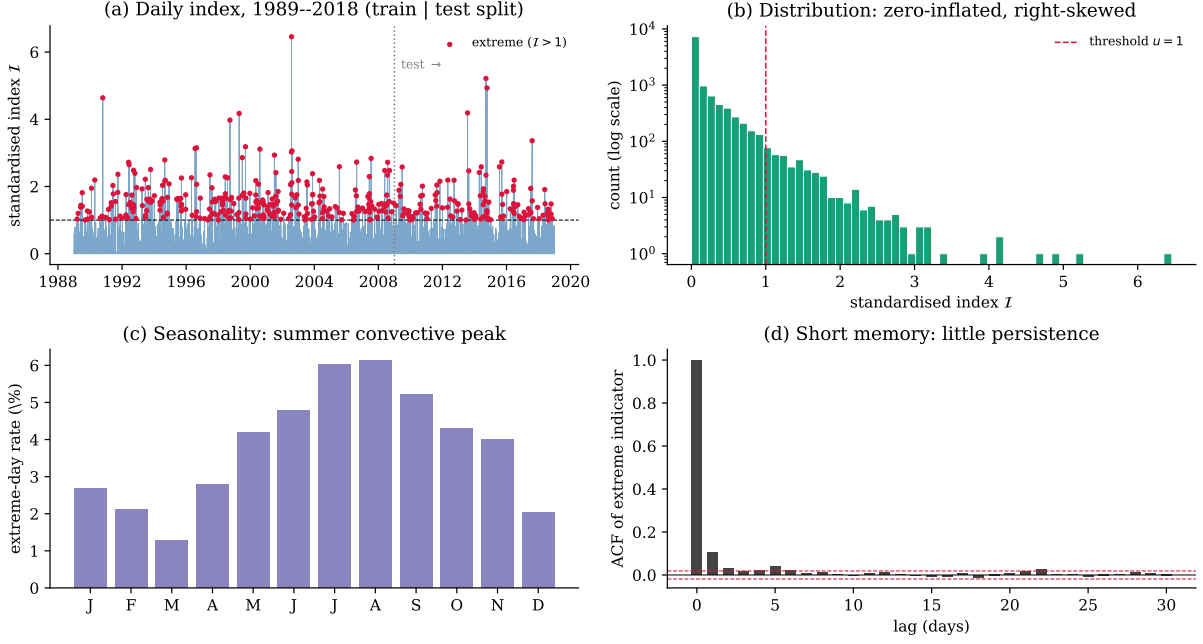


Figure 1: The London 95th-percentile precipitation-extreme index, 1989–2018. **(a)** Daily series with extreme days ( $I > 1$ ) marked and the train/test split; **(b)** the distribution is zero-inflated ( $\sim$ a quarter of days are exactly zero) and strongly right-skewed (log count); **(c)** extremes peak in summer (convective rainfall); **(d)** the extreme indicator has very short memory — the autocorrelation is negligible beyond a day or two (dashed: 95% band) — which caps how much an autoregressive forecaster can achieve.

lagged *drivers*, known strictly before the forecast time. This is the standard perfect-prognosis predictability setting; an operational system would replace antecedent ERA5 with numerical-weather-prediction forecast fields, which we leave to future work (Section ??).

### 3.4 Threshold, training, and testing

We take the **95th-percentile index as the primary target**: its 417 extreme days (287 train, 130 test) admit the full analysis, including the held-out intensity test, which requires extreme days in both training and test across several splits. The rarer 99th-percentile index has only 137 extreme days in thirty years (101 train, 36 test) — too few for the intensity analysis — so its occurrence-only results are given in the Supplementary Material. Unless stated, results are for the 95th percentile at horizon  $h = 1$ .

Because the data are a time series, all splits respect chronology. The primary split trains on 1989–2008 (7,305 days) and holds out 2009–2018 (3,652 days); preprocessing statistics and conformal calibration sets are derived from the training period only. To verify that results are not an artefact of a single test window, the intensity analysis is additionally repeated over four forward-only, expanding-window splits (test periods beginning 2003, 2006, 2009, and 2012), and all stochastic results are averaged over random seeds.

## 4 Methodology

Section ?? fixes notation and the predictability target. Section ?? describes the ladder of models we compare, from climatology and logistic regression through zero-inflated distributional regressions to a deep Gaussian-process extreme-value hurdle. Section ?? describes the conformal calibration we assess (and why its coverage guarantee is, on its own, uninformative). Section ??

sets out the evaluation protocol — proper scores, skill against climatology, a hold-out intensity test, and bootstrap and multi-split robustness — which is itself a contribution.

#### 4.1 Problem set-up and notation

Let  $\mathcal{I}_t \geq 0$  denote the daily standardised precipitation index on day  $t$ , and let  $u > 0$  be a fixed exceedance threshold (we take  $u = 1$ , the value at which the index marks an extreme, and report sensitivity to  $u$ ). A day is *extreme* when  $\mathcal{I}_t > u$ . Write the occurrence indicator and the excess as

$$O_t = \mathbf{1}\{\mathcal{I}_t > u\}, \quad Y_t = (\mathcal{I}_t - u) \mid \{\mathcal{I}_t > u\}, \quad (2)$$

so that  $O_t \in \{0, 1\}$  is observed every day while the excess  $Y_t$  is *defined and observed only on extreme days*. This asymmetry — many occurrence observations, few intensity observations — shapes both the modelling (Section ??) and the separate evaluation of occurrence and intensity predictability (Section ??).

Each day carries a vector of covariates  $\mathbf{x}_t \in \mathbb{R}^d$  that are *strictly antecedent*: functions of information available up to and including day  $t$  (autoregressive lags and rolling summaries of the index, plus lagged ERA5 atmospheric drivers and teleconnection indices; see Section ??). Let  $\mathcal{F}_t$  be the filtration generated by all information up to day  $t$ . For a forecast horizon  $h \geq 1$  the target is the *predictive distribution* of  $\mathcal{I}_{t+h}$  given  $\mathcal{F}_t$ , summarised by its conditional CDF

$$F_{t+h}(z) = \mathbb{P}(\mathcal{I}_{t+h} \leq z \mid \mathcal{F}_t), \quad z \geq 0. \quad (3)$$

Two functionals are of primary interest: the *exceedance probability*  $\pi_{t+h} = \mathbb{P}(\mathcal{I}_{t+h} > u \mid \mathcal{F}_t) = 1 - F_{t+h}(u)$  and the conditional law of the excess  $Y_{t+h} \mid \{\mathcal{I}_{t+h} > u\}$ .

#### 4.2 Models compared

We assess a ladder of models of increasing complexity, all sharing the same strictly antecedent covariates  $\mathbf{x}_t$  and judged by the protocol of Section ??. The aim is not to crown one architecture but to learn *how much* of the index is predictable and *which* modelling choices matter.

**Baselines.** *Climatology* issues the empirical distribution of the training index, the reference against which every skill score is measured; a *stationary peaks-over-threshold* (POT) model adds a Generalised Pareto tail fitted to the training excesses but no covariate dependence.

**Occurrence: logistic regression.** For the exceedance probability  $\pi_{t+h} = \mathbb{P}(\mathcal{I}_{t+h} > u \mid \mathcal{F}_t)$  we fit a regularised logistic regression on  $\mathbf{x}_t$  — the natural calibrated occurrence model.

**Full-distribution zero-inflated regression.** Rather than hurdle at the extreme threshold — which restricts intensity estimation to the few exceedances — we model the *entire* distribution of  $\mathcal{I}$  using all wet days and derive the exceedance probability and tail from it. With a point mass  $p_0(\mathbf{x}_t)$  at zero and a positive continuous CDF  $F_+$ ,

$$F_{t+h}(z) = p_0 + (1 - p_0) F_+(z; \boldsymbol{\theta}(\mathbf{x}_t)) \quad (z > 0), \quad \pi_{t+h} = (1 - p_0)(1 - F_+(u)), \quad (4)$$

with  $p_0$  and the parameters  $\boldsymbol{\theta}$  driven by linear predictors in  $\mathbf{x}_t$ . The Generalised Pareto distribution (GPD) is

$$H(y; \sigma, \xi) = 1 - (1 + \xi y / \sigma)_+^{-1/\xi} \quad (\xi \neq 0), \quad 1 - e^{-y/\sigma} \quad (\xi = 0), \quad y \geq 0, \quad (5)$$

with  $(a)_+ = \max(a, 0)$  and  $\xi$  governing tail heaviness. For  $F_+$  we compare a log-normal, a gamma, and — as the extreme-value-motivated choice — the *extended GPD* (eGPD),

$$F_+(z) = [H(z; \sigma, \xi)]^\kappa, \quad \kappa > 0, \quad (6)$$

which is GPD-tailed at high values (inheriting the Pickands–Balkema–de Haan justification of extreme-value theory; ?) yet is a single continuous model over all positive intensities, requiring *no* threshold selection ( $\kappa$  shapes the lower tail).

**Hurdle with a GPD tail, and its deep-GP variant.** The most elaborate contender is a two-part hurdle whose intensity component is a non-stationary GPD,

$$F_{t+h}(z) = (1 - \pi_{t+h}) + \pi_{t+h} H(z - u; \sigma_{t+h}, \xi), \quad z > u, \quad (7)$$

with  $\pi_{t+h} = \text{sigmoid}(g_\pi(\mathbf{x}_t))$  and  $\sigma_{t+h} = \text{softplus}(g_\sigma(\mathbf{x}_t))$  driven by latent functions on which we place Gaussian-process priors — a deep-kernel GP (a neural feature map composed with a GP kernel; ?) or a deep GP (?) — fitted by sparse variational inference (?). A Gaussian-likelihood GP cannot represent the zero-inflation and heavy tail of  $\mathcal{I}$ , which motivates the EVT likelihood. Full inference details are in Appendix ??.

**Dependent hurdle (a methodological by-product).** The classical hurdle assumes occurrence and intensity are conditionally independent; physically they should be positively correlated, and coupling them lets the abundant occurrence data inform the data-starved tail. We realise this by giving  $(g_\pi, g_\sigma)$  a multi-output GP prior with an intrinsic coregionalisation (??),  $\text{Cov}(g_a(\mathbf{x}), g_b(\mathbf{x}')) = B_{ab} k_\theta(\mathbf{x}, \mathbf{x}')$ , whose off-diagonal yields an interpretable borrowing-strength correlation  $\rho = B_{\pi\sigma} / \sqrt{B_{\pi\pi} B_{\sigma\sigma}}$  ( $\rho = 0$  recovers the independent hurdle). Because the excess is observed only on extreme days  $\rho$  is weakly identified, so we validate recovery on synthetic data (Section ??). This construction — related to correlated two-part models in biostatistics and econometrics (??) — is a by-product, not the paper’s main claim.

**Implementation.** Logistic regression uses  $L_2$  regularisation ( $C = 1$ ); the  $L_1$  feature screen uses  $C = 0.3$ . The distributional regressions place linear predictors on the zero-inflation probability and the positive-part parameters and are fitted by maximum likelihood. The deep-GP–EVT hurdle uses a deep-kernel sparse variational GP ( $M = 64$  inducing points, a two-hidden-layer feature map), with results averaged over five random seeds; conformal intervals target 90% coverage.

### 4.3 Tail-conformal calibration

Any of the models above yields a model-based predictive tail, but under misspecification its quantiles need not be calibrated. As one calibration option we assess, we wrap a model in a conformal layer (??) that supplies distribution-free coverage, with two adaptations for our setting. *Tail targeting:* at the very high confidence levels relevant to extremes the required nonconformity quantile can exceed what a finite calibration set resolves, so we extrapolate the upper tail of the calibration scores with a GPD fit, following the extreme-conformal construction of ?. *Temporal adaptation:* because daily extremes are not exchangeable, we use adaptive conformal inference (?), updating the level online,

$$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t), \quad \text{err}_t = \mathbf{1}\{\mathcal{I}_t \notin C_t\}, \quad (8)$$

with  $C_t$  the issued interval and  $\gamma > 0$  a step size, over a rolling walk-forward window; under (??) with bounded scores the long-run coverage converges to  $1 - \alpha$  irrespective of the data-generating process (?). We stress that this guarantee concerns *coverage*, not predictive quality: as Section ?? shows, a conformally calibrated model can still be worse than climatology on proper scores, so coverage alone is an insufficient indicator and we never report it as evidence of a good forecast.



## 4.4 Evaluation protocol

Because the predictability of extremes is intrinsically limited, evaluation must be both *proper* and *robust*: it must reward calibrated, sharp forecasts (maximising sharpness subject to calibration; ?) and must resist the artefacts — single-run noise, quantile selection — that easily manufacture apparent skill in a low-signal regime. We therefore adopt the following protocol, and treat it as a contribution in its own right.

**Occurrence: Brier decomposition and skill.** For the exceedance probability we report the Brier score, its reliability/resolution/uncertainty decomposition (?) — reliability measuring calibration, resolution measuring informativeness — and the Brier skill score against climatology. A calibrated-but-uninformative forecast (e.g. climatology) has zero resolution, so positive resolution with low reliability is the bar a useful model must clear.

**Whole distribution and tail: (tw)CRPS.** We score the predictive distribution with the continuous ranked probability score and its threshold-weighted form (?),

$$\text{twCRPS}(F_{t+h}, \mathcal{I}_{t+h}) = \int_{\mathbb{R}} (F_{t+h}(z) - \mathbf{1}\{\mathcal{I}_{t+h} \leq z\})^2 w(z) dz, \quad w(z) = \mathbf{1}\{z > \tau\}, \quad (9)$$

which remains proper while emphasising the tail ( $\tau = u$ ). With  $\tau = u$  the sub-threshold distribution is irrelevant, so the tail score depends only on the exceedance probability and the modelled excess.

**Intensity predictability, isolated.** Because the tail-weighted CRPS over all days is dominated by the many non-extreme days, we additionally isolate intensity by scoring the *excess* on observed extreme days: a CRPS of the excess (equivalently, the integral of the pinball loss over quantile levels) relative to the climatological excess distribution. The excess model is a quantile regression whose regularisation is chosen by cross-validation on the training set only, so the test set never informs model selection.

**Robustness.** Skill scores are computed against a climatological reference (the empirical training distribution). Every estimate carries a confidence interval from 1,000–2,000 bootstrap resamples of the test extreme days, and the intensity result is repeated across the four forward-only, expanding-window splits of Section ???. Discrimination metrics (precision–recall AUC) are reported only as secondary context, and conformal coverage is reported but, per Section ??, never as evidence of predictive quality.

## 5 Results

We evaluate every model on the London 95th-percentile index by the proper scores of Section ?? against a climatological reference, with stochastic results seed-averaged and intensity skills carrying bootstrap confidence intervals. Beyond the main bake-off (Section ??) we report an  $L_1$ -regularised feature screen identifying which covariates carry occurrence and intensity signal; the hold-out intensity test across the four expanding-window splits; a seed-averaged horizon sweep  $h \in \{1, 3, 7, 14, 28\}$ ; and a synthetic study, with a known occurrence–intensity correlation, validating the dependent hurdle.

### 5.1 Simpler models win; distribution choice dominates

Table ?? reports the bake-off. Three findings stand out. First, the elaborate deep-GP–EVT hurdle is *worse than climatology* on both the Brier and tail-CRPS skill scores (Brier skill  $\approx -0.4$ ),

Table 2: Model bake-off on London 95th,  $h = 1$  (seed-averaged). Brier skill score (BSS) and tail-CRPS skill (twCRPSS) are relative to climatology; positive is better. RES is Brier resolution (informativeness,  $\times 10^3$ ).

| Model                        | BSS            | RES $\times 10^3$ | twCRPSS        |
|------------------------------|----------------|-------------------|----------------|
| Climatology / stationary POT | 0.000          | 0.0               | 0.000          |
| Logistic (+ERA5)             | <b>+0.038</b>  | 0.83              | —              |
| ZI log-normal (+ERA5)        | −0.29          | —                 | −2.27          |
| ZI gamma (+ERA5)             | +0.008         | 0.79              | −0.06          |
| ZI extended-GPD (+ERA5)      | +0.012         | 0.70              | <b>−0.045</b>  |
| Deep-GP-EVT hurdle (TCDGP)   | $\approx -0.4$ | 0.02              | $\approx -0.3$ |

while a plain calibrated logistic regression and the zero-inflated extended-GPD regression are the best occurrence models. Second, the choice of response distribution matters more than model class: a zero-inflated log-normal is substantially miscalibrated (tail-CRPS skill  $\approx -2$ ), whereas the light-tailed gamma and the threshold-free extended GPD are well behaved — the index tail is lighter than a log-normal or a low-threshold Generalised Pareto fit assumes. Third, modelling the *full* zero-inflated distribution (and deriving the exceedance probability from it) is far better than hurdling at the extreme threshold, which starves the tail of data.

## 5.2 Conformal coverage does not imply forecast quality

The tail-conformal layer attains its nominal 0.90 coverage on the real series across every predictor set and horizon (Figure ??b). But this is *guaranteed by construction* and is uninformative about predictive quality: the same deep-GP-EVT model whose conformal intervals are perfectly calibrated is simultaneously *worse than climatology* on the proper scores (Table ??). The reliability diagram (Figure ??) makes the point directly: the deep-GP-EVT model’s raw exceedance probabilities are badly miscalibrated and almost flat (low resolution), whereas a plain logistic regression is well calibrated and informative. Conformal coverage should therefore not be reported as evidence of a good probabilistic forecast — a cautionary point in its own right.

## 5.3 Predictability of occurrence

Occurrence is weakly but genuinely predictable: the best calibrated model improves on climatology with a Brier skill of +0.038 and positive resolution (Table ??). An  $L_1$ -regularised logistic identifies physically sensible antecedent predictors — led by the **K-index** of convective instability, the 500 hPa geopotential, integrated vapour transport, low-level humidity, and season. Skill nonetheless decays toward the prevalence floor as the forecast horizon grows (Figure ??a), consistent with the limited day-scale persistence of London extremes. At the rarer 99th-percentile threshold, where the scarce data support only an occurrence analysis, the qualitative picture is unchanged (Supplementary Material).

## 5.4 Predictability of intensity

Whether the *magnitude* of an extreme is predictable is the more delicate question, and the answer is a qualified yes. On the extreme days, 23 of 49 antecedent features correlate with the excess at  $p < 0.05$  (versus  $\sim 2$  by chance), led by convective inhibition and CAPE. To test whether this signal generalises, we fit quantile regressions of the excess on held-out extreme days (regularisation chosen by cross-validation on the training set only) and score them by the continuous ranked probability score of the excess against the climatological excess distribution. The CRPS skill is small but its bootstrap interval *excludes zero*, and — crucially — it remains positive across four independent expanding-window splits (Table ??, Figure ??). The signal

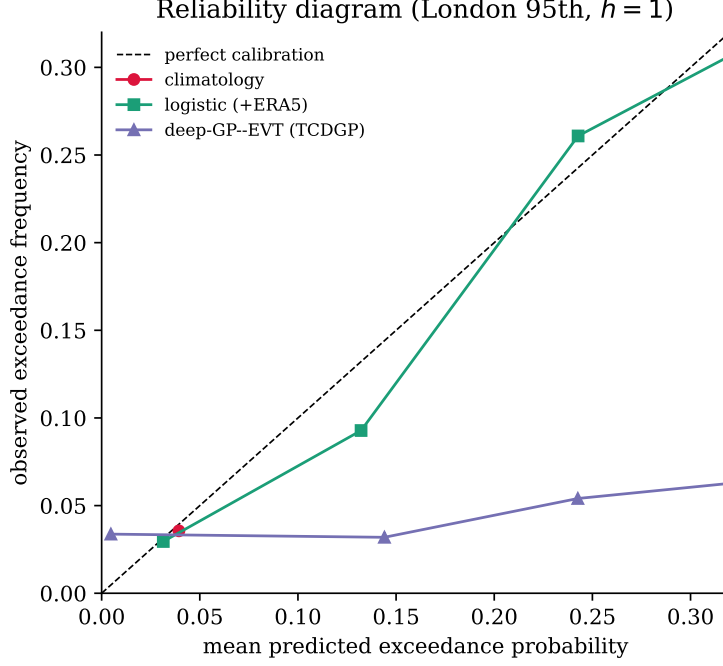


Figure 2: Reliability diagram for the exceedance probability (London 95th,  $h=1$ ). A plain logistic regression tracks the diagonal (calibrated) and spreads its predictions (informative); the deep Gaussian-process extreme-value model is overconfident and nearly flat, despite its conformal intervals attaining nominal coverage. Climatology is a single point at the base rate.

Table 3: Hold-out intensity predictability: CRPS-of-excess skill over climatology across forward-only expanding-window splits, with 90% bootstrap intervals.

| Test period | # test extremes | CRPS skill | 90% CI           | P(> 0) |
|-------------|-----------------|------------|------------------|--------|
| 2003–2018   | 213             | +0.037     | [+0.021, +0.051] | 1.00   |
| 2006–2018   | 175             | +0.027     | [+0.015, +0.041] | 1.00   |
| 2009–2018   | 130             | +0.035     | [+0.021, +0.051] | 1.00   |
| 2012–2018   | 96              | +0.034     | [+0.009, +0.061] | 0.98   |

is concentrated in the upper tail (the largest events) and is carried by antecedent convective instability (CAPE/CIN). Median magnitude, by contrast, is indistinguishable from climatology.

## 5.5 Synthetic validation of the dependent hurdle

On data simulated from the generative model with a *known* occurrence–intensity correlation ( $\rho = 0.70$ , GPD shape  $\xi = 0.20$ ), we verify that the coregionalised hurdle recovers the dependence and that it helps. The *coupled* model recovers  $\hat{\rho} \approx +0.83$ , while the *independent* hurdle (the  $\rho=0$  special case) cannot. Crucially, the independent model’s intensity head — estimated from the few exceedances alone — becomes unstable (its shape collapses so that test excesses fall outside the GPD support), whereas coupling lets the abundant occurrence data stabilise the tail and yields orders-of-magnitude better tail log-likelihood. This is direct evidence for the borrowing-strength mechanism: the dependent hurdle matters most exactly where intensity data are scarcest.

## 5.6 Skill decays with horizon

Both occurrence skill and discrimination decay toward the climatological floor as the forecast horizon grows (Figure ??a), consistent with the limited day-scale persistence of London extremes;

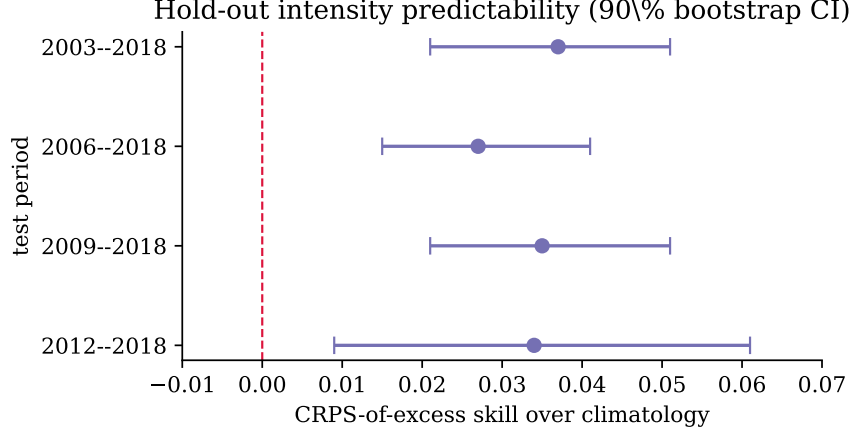


Figure 3: Hold-out intensity predictability: CRPS-of-excess skill over climatology, with 90% bootstrap confidence intervals, across four forward-only expanding-window train/test splits. The skill is small but its interval excludes zero in every split.

conformal coverage, being guaranteed by construction, is flat across horizons (Figure ??b). Seed-averaged sweeps found no statistically robust occurrence-discrimination gain from atmospheric drivers at any single horizon — a reminder that the predictability documented above is weak and easily masked by noise unless assessed with seed-averaging and proper scores rather than single runs.

**Summary.** Across the bake-off, simpler calibrated models dominate the elaborate deep-GP-EVT model, and the response distribution matters more than the model class (Section ??); conformal coverage holds but is, on its own, uninformative about forecast quality (Section ??). The substantive finding is a small but *rigorously established* predictability: occurrence is weakly predictable from antecedent convective-instability and circulation indices (Section ??), and the magnitude of the largest events is weakly predictable from antecedent convective instability, with a CRPS skill over climatology that excludes zero across four independent splits (Section ??). The dependent-hurdle mechanism is separately validated on synthetic data (Section ??).

## 6 Discussion

**Predictability is real but small, and physically interpretable.** The headline is not a forecasting triumph but an honest measurement: for London, the occurrence of a daily precipitation extreme and the magnitude of the largest events both carry weak antecedent predictability, on the order of a few percent of CRPS skill over climatology. That this signal is led by convective-instability indices (the K-index for occurrence; CAPE and convective inhibition for the intensity of the largest events) is reassuring: it is the thermodynamics one would expect to govern intense convective rainfall, and it argues against the skill being an artefact. The median magnitude of an extreme, by contrast, is indistinguishable from climatology: once an exceedance occurs, its size is largely unpredictable except in the upper tail.

**Methodological lessons.** Three are worth emphasising. First, *model complexity did not pay*: a deep Gaussian-process extreme-value hurdle was beaten by logistic regression and by simple zero-inflated distributional regressions. Gaussian processes earn their keep through structure (notably spatial structure), and a single-site temporal problem offers them little to exploit. Second, the *response distribution dominates*: a log-normal tail was substantially miscalibrated, and hurdling at the extreme threshold starved the tail of data; modelling the full zero-inflated

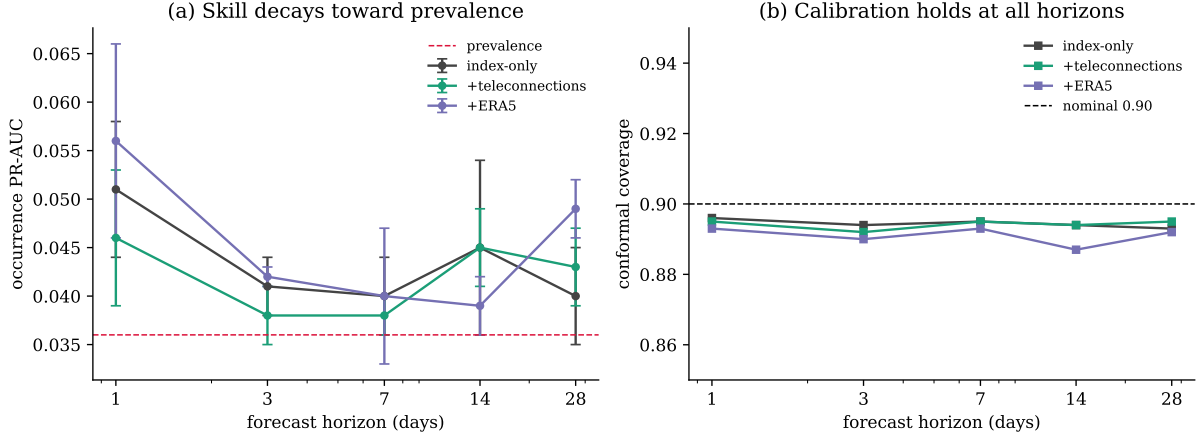


Figure 4: Seed-averaged horizon sweep on the London 95th-percentile index. **(a)** Occurrence PR-AUC decays toward the prevalence floor (dashed) as the forecast horizon grows; driver gains are marginal at all horizons (the apparent ERA5 edge at  $h = 28$  does not survive a higher-seed check; see text). **(b)** Conformal coverage stays at the nominal 0.90 across all horizons and predictor sets. Error bars / markers are means over three seeds.

distribution with a threshold-free extended GPD was far better. Third, and most importantly for the literature, *conformal coverage is insufficient as a stand-alone indicator of forecast quality*: it is attained by construction even by a model that is worse than climatology on proper scores. Reporting coverage without a proper-score comparison can therefore create a false impression of a good probabilistic forecast.

**Why rigour mattered here.** Several apparent results did not survive scrutiny: a single-seed driver benefit and an apparent long-horizon gain both evaporated under seed-averaging, and an across-the-board intensity skill shrank to an upper-tail effect once regularisation was chosen without peeking at the test set. The predictability we report is what remained after seed-averaging, proper scoring, cross-validated model selection, bootstrap intervals, and multi-split robustness. In a low-signal regime these safeguards are not optional — they are the difference between a finding and an artefact.

**Limitations.** The study is single-site and uses a perfect-prognosis design (antecedent reanalysis fields, not operational forecasts); the intensity analysis conditions on observed exceedances and so isolates predictability rather than delivering an end-to-end forecast; and the index is itself an ERA5-derived product, which we mitigate by excluding all precipitation fields and lagging every predictor.

## 7 Conclusion

We set out to measure, rather than merely to model, the predictability of rare precipitation extremes in London, and to do so with an evaluation protocol robust to the pitfalls of a low-signal setting. The answer is a qualified, quantified “a little”: occurrence and the magnitude of the largest events are weakly but genuinely predictable from antecedent convective instability, while simpler calibrated models suffice and conformal coverage is no substitute for a proper-score comparison. Natural extensions are the directions in which more signal — and a genuine role for structured Gaussian-process models — is most plausible: *spatial* modelling across the grid of cells and across European cities, *operational* drivers from numerical weather prediction in place of reanalysis, and *sub-seasonal* aggregations where slowly varying drivers may contribute.

Whether predictability rises materially in those settings is, in the spirit of this paper, a question to be settled by proper scores and confidence intervals rather than asserted.

## Data and code availability

The underlying data are publicly available: the London precipitation-extreme index derives from the Copernicus Climate Data Store product *Extreme precipitation risk indicators for Europe and European cities (1950–2019)*, and the atmospheric predictors from ERA5 (Copernicus CDS, `reanalysis-era5-single-levels`; and the WeatherBench2 regridded ERA5 archive) and from the NOAA teleconnection indices. The code accompanying this paper — implementations of all models (logistic, the zero-inflated log-normal/gamma/eGPD distributional regressions, and the deep-GP-EVT hurdle), the full evaluation protocol (proper scores, the hold-out intensity test, bootstrap and multi-split robustness), the data-retrieval and feature-processing scripts, and the scripts that produce every figure and table — together with the processed daily feature table, will be deposited in a public repository ([REPOSITORY URL / ARCHIVED DOI TO BE INSERTED ON ACCEPTANCE]).

## A Data and predictors

**Response.** The standardised daily precipitation-extreme index for London (Section ??), derived from the ERA5-2km downscaled reanalysis; we use the daily maximum over the 53 London grid cells, with an extreme defined by  $\mathcal{I} > 1$ , at the 95th (primary) and 99th percentile standardisations.

**Antecedent predictors.** All predictors are functions of information up to the forecast origin  $t$  (a horizon- $h$  forecast of day  $t + h$  uses day- $t$  information only); no precipitation field is used, since the response is itself an ERA5-derived precipitation product.

- **Index history:** lags  $\mathcal{I}_{t-1}, \dots, \mathcal{I}_{t-14}$ ; rolling mean over the previous 7 and 30 days; rolling maximum over 7 days; count of exceedances in the previous 30 days; and sin / cos harmonics of day-of-year.
- **ERA5 circulation and moisture** (WeatherBench2,  $1.5^\circ$ , daily means over a London-centred box,  $45\text{--}60^\circ\text{N}$ ,  $20^\circ\text{W--}10^\circ\text{E}$ ): mean sea-level pressure and its meridional/zonal gradients, total-column water vapour, 2m temperature, 10m wind components and speed, geopotential at 500 and 850 hPa, and specific humidity at 850 hPa.
- **ERA5 convective instability** (Copernicus CDS, `reanalysis-era5-single-levels`,  $0.25^\circ$ , daily mean and maximum over the box): CAPE, convective inhibition (CIN), the K-index, the Total-Totals index, total-column water vapour, the vertical-integral eastward/northward water-vapour-flux components and their magnitude (IVT).
- **Teleconnection indices** (monthly, broadcast to daily): North Atlantic Oscillation and Arctic Oscillation (NOAA CPC), and Niño 3.4 (NOAA PSL).

**Retrieval.** WeatherBench2 fields were read directly from the public Google Cloud zarr store with `xarray`; the convective-instability fields were retrieved from the Copernicus CDS via `cdsapi` with server-side spatial subsetting to the box, 6-hourly sampling aggregated to daily statistics; teleconnection indices were downloaded from NOAA. Every driver is forward-filled to daily resolution where needed and lagged by the forecast horizon. Processing code and the cached daily feature table accompany the paper.

## B The deep-GP extreme-value hurdle: inference

We give the inference for the most elaborate contender (Section ??). The parameters depend on covariates through latent functions,  $\pi = \text{sigmoid}(g_\pi(\mathbf{x}_t))$ ,  $\sigma = \text{softplus}(g_\sigma(\mathbf{x}_t))$ , and a constant shape  $\xi_0$  (covariate-dependent  $\xi$  is an optional extension). For horizon  $h$  the per-day likelihood factorises into occurrence and (on extreme days) intensity terms,

$$p(O_t, \mathcal{I}_t \mid \mathbf{g}(\mathbf{x}_{t-h})) = \pi^{O_t} (1 - \pi)^{1-O_t} [h_{\text{GPD}}(\mathcal{I}_t - u; \sigma, \xi_0)]^{O_t} [f^{\text{blw}}(\mathcal{I}_t)]^{1-O_t}, \quad (10)$$

with  $h_{\text{GPD}}$  the GPD density and  $f^{\text{blw}}$  a zero-inflated positive density for the sub-threshold regime.

**Sparse variational inference.** We place GP priors on  $\mathbf{g} = (g_\pi, g_\sigma)$  and introduce  $M \ll N$  inducing variables  $\mathbf{u} = \mathbf{g}(\mathbf{Z})$  with whitened variational posterior  $q(\mathbf{u}) = \mathcal{N}(\mathbf{m}, \mathbf{S})$ . The evidence lower bound is

$$\mathcal{L} = \sum_t \mathbb{E}_{q(\mathbf{g}(\mathbf{x}_{t-h}))} [\log p(O_t, \mathcal{I}_t \mid \mathbf{g}(\mathbf{x}_{t-h}))] - \text{KL}(q(\mathbf{u}) \parallel p(\mathbf{u})), \quad (11)$$

the expectation computed by Monte Carlo over the marginal  $q(\mathbf{g}(\mathbf{x}_{t-h}))$ . Inducing-point inference reduces cost to  $\mathcal{O}(NM^2)$ ; the kernel acts on a neural feature map (deep kernel) or, for the deep GP, on a composition of GP layers fitted by doubly-stochastic variational inference (?). Hyperparameters, inducing inputs/distribution, neural weights, and  $\xi_0$  are learned jointly by maximising (??).

**Coregionalised (dependent) hurdle.** We couple the two latents through a linear model of coregionalisation: with  $Q$  shared base GPs  $f_q \sim \mathcal{GP}(0, k_\theta)$ ,  $g_a(\mathbf{x}) = \sum_q W_{aq} f_q(\mathbf{x})$ , so  $\text{Cov}(g_a, g_b) = B_{ab} k_\theta$  with  $\mathbf{B} = \mathbf{W}\mathbf{W}^\top$ . The correlation  $\rho = B_{\pi\sigma} / \sqrt{B_{\pi\pi} B_{\sigma\sigma}}$  is the borrowing-strength parameter; a diagonal  $\mathbf{W}$  recovers the independent hurdle. Because the excess is observed only on extreme days,  $\rho$  is weakly identified, so we use a weakly-informative prior shrinking  $\rho \rightarrow 0$  and validate recovery on synthetic data (Section ??).

**Tail-conformal layer.** For an interval at level  $1 - \alpha$  we use a conformalised-quantile-regression nonconformity score on the model’s predictive quantiles,  $E_t = \max\{\hat{z}_{\alpha/2} - \mathcal{I}_t, \mathcal{I}_t - \hat{z}_{1-\alpha/2}\}$ ; when the required score quantile exceeds what the calibration set resolves we extrapolate with a GPD fit to the upper calibration scores (?), and we adapt the level online by (??) over a rolling window.

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